CHAPTER 14

Computer Network

Learning Objectives

By the end of this chapter, the students will be able

- To explain the evolution of Networking
- Types of Network Topologies
- To explain the types of Network
- To compare the types of Networks
- To identify the types of network in the lab
- To identify computers and users over a network
- To explain wireless mobile communication
- To explain internet applications
- To know Network security concepts
- To use a DNS server to connect it to a network resource/server

Computer Network

A computer network is an interconnection of various computers to share software, hardware, resources and data through a communication medium between them.

The computers on a computer network may be linked through cables, telephone lines, radio waves, satellites, or infrared light beams.

14.1. Evolution of Networking

Computers exchange data in automatic mode, is, essentially, the basic mechanism of any computer network. Developers of the first networks implemented services for file exchange, database synchronization, e-mail and other network services.

14.1.1. ARPANET

ARPANET was established by the Advanced Research Projects Agency (ARPA) in 1969 for two main reasons:

- To allow the transfer of data between various research institutes.
- To answer the call of the U.S. Department of Defence for a technology to provide messaging capabilities to the government in the event of nuclear war.

14.1.2 World Wide Web(WWW)

The World-Wide Web is a collection of documents and services. It is distributed across the Internet and linked together by hypertext links. The web is therefore a subset of the Internet.

World Wide Web was created by **Timothy Berners Lee** in 1989 at **CERN** in **Geneva**.

Web page is a document available on World Wide Web. A web page can contain